Project Planning & Management

1- Project Proposal

Project Name
Beyond the Dose💊

Problem Statement
The growing use of GLP-1 medications for weight management has generated diverse data on reported adverse events, clinical trials, public interest, and stock market performance. Analyzing these data sources can reveal important trends that are difficult to identify when examined separately.
This project aims to analyze these trends using Excel and Power BI, transforming available data into meaningful insights through interactive dashboards.

Project Objectives
1- Identify the most frequently reported adverse reactions and serious outcomes.
2- Analyze report trends by time, age, sex, and country.
3- Explore clinical trial phases, statuses, and sponsors.
4- Compare stock performance and public search interest.
5- Present findings through interactive Power BI dashboards.

Dataset Description & Source
The project will use the **GLP-1 Weight Loss Drugs Master Dataset (2017–2026),** available on **Kaggle**.
https://www.kaggle.com/datasets/devtayyabsajjad/glp-1-weight-loss-drugs-master-dataset-2017-2026
The dataset brings together multiple data sources, including FDA adverse event reports, clinical trial information, stock market data, and public search interest.These sources will be prepared for analysis to identify relevant trends and relationships.

Tools & Technologies
Excel: Data exploration, initial cleaning, and preliminary analysis.
SQL: Querying, filtering, and aggregating data across tables.
Power Query: Data cleaning, transformation, and preparation for analysis.
Power BI: Data modeling, KPI analysis, and interactive dashboard creation.

2- Project Plan

Stage	Start	End	Duration
Understanding data	16 OCT	20 OCT	5 Days
Data Cleaning and Preparation	21 OCT	28 OCT	8 Days
Analysis using Excel and SQL	29 OCT	5 NOV	8 Days
Implementing Power BI dashboard	6 NOV	15 NOV	10 Days
Reviewing results	16 NOV	20 NOV	5 Days
Documentation & Final Presentation	21 NOV	30 NOV	10 Days

3- Task Assignment & Roles

Members	Role	Responsibilities
Hana	Project Planning	Organizing the project and defining the business questions.
Fatma & Mai	Data Quality & Cleaning	Inspecting, cleaning, and preparing the dataset using Excel.
Rahma & Esraa	SQL Data Analysis	Querying, filtering, and analyzing data using SQL.
Hana, Rahma & Esraa	Power BI Dashboard Development	Building interactive dashboards and visualizing key findings using Power BI.
All	Results Review	Reviewing results, validating findings, and documenting the project.

4- Risk Assessment & Mitigation plan

Missing or duplicate data

Potential Impact:
Inaccurate analysis and misleading results.
Mitigation Plan:
Inspect data quality, identify missing and duplicate values, and handle them appropriately.

Diversity of data type

Potential Impact:
Errors in calculations and charts.
Mitigation Plan:
Configuring column types in power query.

Errors in calculations or visualizations

Potential Impact:
Incorrect KPIs and misleading dashboard insights.
Mitigation Plan:
Cross-check calculations, validate results against source data, and test dashboard filters.

Large dataset size

Potential Impact:
Slow processing and analysis.
Mitigation Plan:
Focus on relevant columns, optimize queries, and analyze data in manageable stages.

Time constraints or the unavailability of a team member

Potential Impact:
Delay in project delivery.
Mitigation Plan:
Weekly follow-up, documentation of work, and file sharing.

5- Key Performance Indicators (KPIs)

Total Unique Adverse Event Reports
Importance: Measures the total volume of reported adverse events without double-counting the same report.
Calculation: Count of distinct report IDs.

Serious Report Rate
Importance: Helps compare the proportion of serious reports across different drugs.
Calculation: (Unique Serious Reports ÷ Unique Reports with Known Seriousness Classification) × 100.

Death Report Rate
Importance: Helps compare the proportion of reports involving death across drugs.
Calculation: (Unique Reports Involving Death ÷ Total Unique Reports) × 100.

Hospitalization Report Rate
Importance: Helps compare hospitalization-related reports across drugs.
Calculation: (Unique Reports Involving Hospitalization ÷ Total Unique Reports) × 100.

Clinical Trial Count
Importance: Number of clinical trials by phase, status, or sponsor.
Calculation: Count of distinct clinical trial IDs for each category.

Number of Unique Drugs Analyzed
Importance: Shows the range of medications covered by the project and helps compare adverse event reports across drugs.
Calculation: Count of distinct drug names.